DAY 4 · FRIDAY 30 OCTOBER 2026 · MFB 545

Blockchain, crypto-assets, DeFi and tokenisation

A blockchain is a ledger that nobody keeps alone and everybody can verify. This day shows it for real: you build a blockchain among yourselves, then you read on DefiLlama how much money decentralised finance carries today, where, and what it loses. Then you borrow and swap on protocols with no bank, you split a building into tokens, you see how hundreds of millions get stolen — and you decide whether your team's pension-savings app needs a blockchain, or only a proof that the money is there.

🕘 **09:00–12:15 · 13:30–16:45 · four 90-minute blocks**

💻 **laptop · an up-to-date Chrome, Firefox or Brave**

🎤 **the talk is short — 🛠️ most of the time, you are the one working**

THE DAY'S WORKSHOP · THE CLASS BLOCKCHAIN

Everyone in the room is a node: make your keys, sign, hash, mine, pay someone, then cheat — and see why cheating fails. Section 4.1 B, from the first block.

⛓️ **Open the class blockchain ↗** · 🖥️ **The room screen ↗**

The brief · section 4.1 B

The afternoon: blockchains and DeFi in numbers, on DefiLlama — a slide deck (in French), then the section 4.2 workshop, which reads the same data live.

🧱 **The slides ↗** · **The workshop · section 4.2**

BY THE END OF DAY 4, YOU WILL BE ABLE TO

- Explain in three ideas why a shared ledger holds without a trusted third party — signature, fingerprint, consensus — and what it does not guarantee.
- Explain, having done it among yourselves, why a shared ledger holds without a keeper: keys, hashes, mining, consensus — and what it does not prove.

- Read DefiLlama: TVL, the chains that matter, the protocol families, the weight of stablecoins, who earns the fees and how billions get lost.
- Calculate the health factor of a DeFi loan, its liquidation price, and the price obtained on an automated exchange.
- Tell apart the three families of stablecoins and say what backs each one.
- Place tokenisation and central bank digital currencies in Switzerland (DLT Act, SDX, Helvetia) and in WAEMU.
- Decide, with a published method, whether your project needs a blockchain — and prove to a customer that their units exist.

BEFORE YOU START — TICK

- ☑ **Signed in** on this site (the button at the bottom right).
- ☑ **An up-to-date browser** (Chrome, Firefox or Brave): the class blockchain and DefiLlama run in it, nothing to install.
- ☑ **A calculator** (the phone's is enough): shares, health factors and swap prices are worked out by hand before the page checks them.
- ☑ **Your team's pension-savings plan** (Day 3, section 3.9): the day ends on it.

BLOCK 1 · 09H00–10H30 why "trustless" works · build your blockchain · the DefiLlama tour

4.1 **Why "trustless" works**
Signature, fingerprint, consensus · the life of a transaction · what the chain guarantees
🎤 TALK · 20 MIN

4.1 B **Build your blockchain**
Every student is a node · keys, hashes, mining, payments, forks, cheating · what the ledger proves
🛠 WORKSHOP · 60 MIN

4.2 **The DefiLlama tour: decentralised finance in numbers**
Live TVL · chains and protocols · stablecoins · fees, revenue, hacks

🛠 WORKSHOP · 55 MIN

BLOCK 2 · 10:45–12:15 lend, swap, stabilise · the DeFi labs · the quiz

4.3 **Lend, swap, stabilise — with no bank**

Health factor and liquidation · x × y = k · three families of stablecoins

🎤 TALK · 15 MIN

4.4 **DeFi labs · borrow, swap**

The loan and its liquidation price · price impact · the DeFi capstone

🛠 WORKSHOP · 50 MIN

4.5 **End-of-morning quiz**

Eight questions · launched in class · counts towards participation

✅ QUIZ · 10 MIN

BLOCK 3 · 13:30–15:00 tokenisation, Switzerland and WAEMU · “do you need a blockchain?”

4.6 **Tokenisation and central bank digital currencies**

A token, a right · DLT Act, SDX, Helvetia · the BCEAO and the eNaira

🎤 TALK · 20 MIN

4.7 **Split a building · do you need a blockchain?**

The fractionalisation simulator · the decision tree applied to your app

🛠 WORKSHOP · 55 MIN

BLOCK 4 · 15:15–16:45 MEV, hacks, MiCA · prove the money is there · DeFi Millionaire

4.8 **MEV, three hacks, MiCA**

The sandwich attack in numbers · The DAO, Ronin, Bybit · MiCA, FINMA, BCEAO

🎤 TALK · 15 MIN

4.9 **Prove the money is there · DeFi Millionaire**

The customers' Merkle tree · your app's proof and keys · the game

🛠 WORKSHOP · 75 MIN

← Course page

reset my Day 4 progress on this device

SECTION 4.1 · BLOCK 1

🎤 TALK · 20 MIN

Why “trustless” works

The instructor’s talk. If it does not take place, read this section on your own, then do the sorting at the bottom of the page.

THE PROBLEM

A ledger that nobody keeps alone

Your bank account is a line in the bank’s ledger. You believe the balance because you trust the bank, its auditor and FINMA. A blockchain replaces that third party with a ledger **copied on thousands of computers**, which anyone can reread from day one. Three questions must then be answered with no one in charge: who may spend? can the past be rewritten? in what order do transactions go through? Three ideas, one per question.

- **1 · The signature — who can spend.** Each account is a pair of keys. The **private** key signs; the **public** key (from which the address comes) lets everyone check the signature without ever knowing the private key. Without a valid signature, no computer on the network accepts the transaction. That is the sheet public key and private key.
- **2 · The fingerprint — the past is not rewritten.** Each block contains the **fingerprint** (the hash) of the previous block. Changing one comma in an old block changes its fingerprint, and therefore that of every block after it: the forgery shows immediately. That is the sheet what is a hash?.
- **3 · Consensus — who writes the next block.** Bitcoin draws lots by computation (*proof of work*, one block every ten minutes or so). Ethereum, since September 2022, by stake (*proof of stake*): validators each lock up 32 ETH; every 12 seconds one of them proposes a block and the others attest it. A validator who cheats loses part of its stake. After two “epochs” (about 13 minutes) the block is **finalised**: undoing it would cost at least a third of all the stake.

These three ideas are enough. The rest — smart contracts, DeFi, tokens — is code that runs on this ledger and inherits its guarantees, *and its limits*.

THE MECHANISM · THE LIFE OF A TRANSACTION

Seven steps between “Send” and the public ledger

▸ 0.001 ETH SENT TO A CLASSMATE ON ETHEREUM

1 · The wallet builds the transaction. Recipient, amount, a counter (the *nonce*: this is your 1st, 2nd… transaction, which prevents it being replayed), and two caps: the maximum **gas** (the computing work allowed; a simple transfer uses exactly 21,000) and the maximum price per unit of gas.

2 · It signs it with the private key. The key never leaves your device; only the signature goes out. Anyone can check, with your public key, that it really is you and that nothing has been altered.

3 · It goes into the public queue. The *mempool*: thousands of nodes receive it and check the signature and the balance. It is visible to everyone *before* it is executed — remember this for section 4.8.

4 · A validator picks it. Every 12 seconds, a validator drawn by lot proposes a block; it orders the transactions, usually those offering the best tip first.

5 · It is executed and paid for. The balance changes. Fee = gas used × gas price. Since 2021 (EIP-1559), the "base" part of that price is **destroyed**; only the tip goes to the validator.

6 · It becomes final. The other validators attest the block; after two epochs, about 13 minutes, it is **finalised**. No customer service can cancel it or refund it.

7 · Anyone can read it. A block explorer (Etherscan) is just a website that reads the public ledger: sender, recipient, amount, fee, block, time. Your name is not there — but your address, once linked to you, reveals your whole history.

✓ All 7 steps are shown

CHECK · 5 MIN

What the blockchain guarantees — and what it does not

› GUARANTEED BY THE CHAIN · NOT GUARANTEED: SOMEONE ELSE IS NEEDED

That the transaction was authorised by the holder of the private key

Guaranteed by the chain | Not guaranteed

That a finalised transaction will not be altered

Guaranteed by the chain | Not guaranteed

That the building represented by a token exists and is not mortgaged twice

Guaranteed by the chain | Not guaranteed

That a smart contract does what is written in its code

Guaranteed by the chain | Not guaranteed

That an address really belongs to the person who claims it

Guaranteed by the chain | Not guaranteed

That funds can be recovered after losing the secret phrase

Guaranteed by the chain | Not guaranteed

That the balance shown by a stablecoin is covered by dollars at the bank

Guaranteed by the chain | Not guaranteed

0 of 7 sorted

SECTION 4.1 B · BLOCK 1

🛠 WORKSHOP · 60 MIN

Build your blockchain

A · 45 MIN · THE CLASS BLOCKCHAIN

Everyone in the room is a node

Section 4.1 says why a shared ledger holds without a trusted third party. Before looking at the real ones (section 4.2), build one among yourselves: your browser keeps its own copy, checks every block and every payment, and refuses what is false. The site only carries the messages. Seven rounds: keys and signature, hash and block, mining, payments, the chain, cheating, the project.

⛓ Open the class blockchain ↗ 🖥 The room screen ↗

▸ WHAT THE WORKSHOP RECORDED ON THIS DEVICE

- ☐ Key pair made (round 1)
- ☐ Message signed and verified (round 1)
- ☐ A nonce found by hand (round 2)
- ☐ A block mined on the class chain (round 3)
- ☐ A payment signed and sent (round 4)
- ☐ My copy altered — and refused by the others (round 6 A)
- ☐ A double spend attempted (round 6 B)
- ☐ My copy repaired

The workshop page opens in its own tab; this list updates when you come back here.

B · 15 MIN · FOR THE PROJECT

Which entry of your app deserves this ledger?

What you just built guarantees three things: a signed entry cannot be forged on the customer’s behalf; a past entry cannot be rewritten without everyone seeing it; no single

keeper can alter the ledger. It does not guarantee that the units exist at the custodian — the chain proves the entry, not the real world (section 4.9).

✍️ **One entry of your app that would gain from a shared ledger (which one, and who the nodes would be); one that does not need it (and why an ordinary database is enough); and what the ledger will never prove.**

e.g. Each customer's unit statement: nodes = our company, the Swiss custodian, an auditor, the regulator…

DEBRIEF · 10 MIN

On the room screen: the nodes, the longest chain, the balances. Who won the most blocks, and why? What happened to the payment spent twice? And when the instructor altered the site's copy, what did your nodes do?

SECTION 4.2 · BLOC 1

🛠️ ATELIER · 55 MIN

The DefiLlama tour: decentralised finance in numbers

THE WORKSHOP RULE

DefiLlama ↗ is the public dashboard of decentralised finance: free, no account, no ads, open source. It reads the contracts of more than 400 blockchains and adds up what they hold. This page reads the same data (live) and asks you questions about it; open DefiLlama alongside and check every figure at the source.

Two words first. **TVL** (*total value locked*) is the value, in dollars, of the tokens deposited in the contracts of a protocol or a chain: the money at work, not the money changing hands. A **protocol** is a program deployed on a chain (Aave, Uniswap, Lido); a **chain** is the ledger that carries it (Ethereum, Solana, Tron).

Open DefiLlama ↗

🧱 The slides: blockchains and DeFi in numbers (French) ↗️

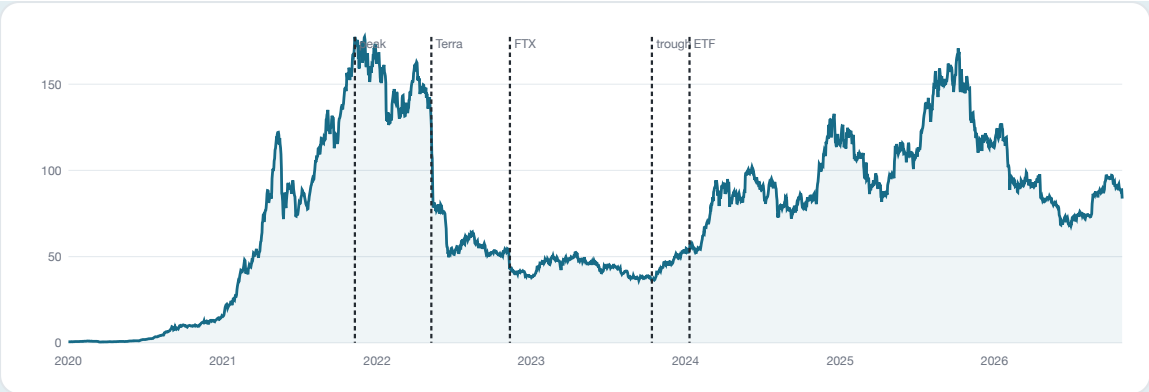
A · 10 MIN · THE MAP

How much money, where, and since when

83.9 Md $	420	5,489	51.2 %
TVL OF ALL DEFI	CHAINS TRACKED	DEFI PROTOCOLS (30 OCT)	ON ETHEREUM

77,556 $	2,242 $
ONE BITCOIN	ONE ETHER

▸ THE TVL OF ALL DEFI SINCE 2020, IN BILLIONS OF DOLLARS



Move the mouse over the curve. The marks: the peak of November 2021, the Terra collapse (May 2022), the FTX bankruptcy (November 2022), the trough of October 2023, the US bitcoin ETFs (January 2024).

What share of total TVL is on **Ethereum**, in % (read the two figures above)?

… % Check

How far did TVL fall between the **peak** (2021-11-09, 177.5 bn) and the **trough** that followed (2023-10-13, 36.0 bn), in %?

… % Check

B · 15 MIN · CHAINS AND PROTOCOLS

Who carries the money, and who puts it to work

▸ THE FIFTEEN LARGEST CHAINS BY TVL

1 · Ethereum		43.0 Md · 51.2 %
2 · Bitcoin		6.0 Md · 7.1 %
3 · Base		5.8 Md · 6.9 %
4 · Tron		5.6 Md · 6.6 %
5 · Solana		5.4 Md · 6.5 %
6 · BSC		5.2 Md · 6.3 %
7 · Hyperliquid L1		2.7 Md · 3.2 %
8 · Arbitrum		1.4 Md · 1.7 %
9 · Robinhood Chain		1.1 Md · 1.3 %
10 · Monad		0.9 Md · 1.0 %
11 · Polygon		0.7 Md · 0.8 %

12 · Arc		0.6 Md · 0.7 %
13 · Avalanche		0.5 Md · 0.6 %
14 · Plasma		0.5 Md · 0.6 %
15 · Sui		0.5 Md · 0.5 %

The first 1 chain(s) carry 51.2 % of TVL · slide

› THE TWENTY LARGEST DEFI PROTOCOLS, CENTRALISED EXCHANGES EXCLUDED · SNAPSHOT OF 30 OCTOBER 2026

#	PROTOCOL	CATEGORY	CHAINS	TVL
1	**Lido**	Liquid staking	5	21.7 Md $
2	**LayerZero V2**	Bridge	91	15.7 Md $
3	**Aave V3**	Lending	21	14.6 Md $
4	**SSV Network**	Staking pool	1	11.7 Md $
5	**Morpho Blue**	Lending	45	10.5 Md $
6	**WBTC**	Bridge	1	9.0 Md $
7	**Binance staked ETH**	Liquid staking	2	8.6 Md $
8	**Coinbase Bridge**	Bridge	5	7.7 Md $
9	**Hyperliquid Bridge**	Bridge	2	6.6 Md $
10	**EigenCloud**	Restaking	1	5.8 Md $
11	**Binance Bitcoin**	Bridge	1	5.3 Md $
12	**Sky Lending**	Collateralised stablecoin (CDP)	1	5.1 Md $
13	**SparkLend**	Lending	2	5.1 Md $
14	**Ethena USDe**	Basis trading	1	4.5 Md $
15	**ether.fi Stake**	Liquid staking	6	4.3 Md $
16	**JustLend V1**	Lending	1	3.6 Md $
17	**Arbitrum Bridge**	Canonical bridge	1	3.3 Md $
18	**JustLend sTRX**	Liquid staking	1	3.3 Md $

19	**Babylon Protocol**	Restaking	1	3.2 Md $
20	**USDT0**	Bridge	1	3.1 Md $

All DeFi protocols together: 262.8 bn $ (some is counted twice when one protocol deposits into another). 4,276 protocols weigh under one million; 51 exceed one billion. DefiLlama also ranks the reserves of centralised exchanges (Binance, OKX…); they are left out here: a centralised exchange is not decentralised finance.

How many chains, starting from the first, carry **90%** of TVL? (the slider above)

… chains Check

The **top ten protocols** in the table weigh what share of all protocols' TVL (262.8 bn $), in %?

… % Check

› WHICH FAMILY DOES EACH PROTOCOL BELONG TO?

Aave V3

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

Lido

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

Uniswap

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

LayerZero

Lending | Liquid staking | Exchange (AMM) | Bridge

Collateralised stablecoin

Sky (ex-MakerDAO)

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

Morpho

Lending | Liquid staking | Exchange (AMM) | Bridge | Collateralised stablecoin

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C · 10 MIN · STABLECOINS

Three hundred billion dollars that are not dollars

300.8 Md $	334	59.9 %	91.1 %
ALL STABLECOINS	STABLECOINS TRACKED	USDT (TETHER)	BACKED BY RESERVES

3.6 ×
DEFI'S TVL

▸ THE TEN LARGEST STABLECOINS, WITH THEIR CURRENT PRICE

#	TOKEN	BACKING	CIRCULATING	SHARE	PRICE
1	**Tether** · USDT	fiat-backed	180.34 Md $	59.9 %	0.9992 $
2	**USD Coin** · USDC	fiat-backed	71.73 Md $	23.8 %	0.9997 $
3	**Sky Dollar** · USDS	crypto-backed	6.51 Md $	2.2 %	1.0014 $
4	**Dai** · DAI	crypto-backed	4.74 Md $	1.6 %	1.0002 $

5	**Ethena USDe** · USDe	crypto-backed	4.53 Md $	1.5 %	0.9987 $
6	**World Liberty Financial USD** · USD1	fiat-backed	3.89 Md $	1.3 %	0.9982 $
7	**Global Dollar** · USDG	fiat-backed	2.99 Md $	1.0 %	1.0003 $
8	**PayPal USD** · PYUSD	fiat-backed	2.96 Md $	1.0 %	1.0010 $
9	**Ripple USD** · RLUSD	fiat-backed	2.56 Md $	0.9 %	1.0002 $
10	**BlackRock USD** · BUIDL	fiat-backed	2.10 Md $	0.7 %	1.0000 $

None of the ten is more than half a cent from the dollar right now. March 2023: USDC fell to $0.88 over a weekend, because 3.3 billion of its reserves sat at Silicon Valley Bank. May 2022: Terra's UST fell to zero.

USDT and USDC together: what share of all stablecoins, in %?

… % Check

D · 10 MIN · WHAT IT EARNS, WHAT IT LOSES

Fees, revenue and hacks

› FEES PAID BY USERS AND REVENUE KEPT BY THE PROTOCOL, OVER 30 DAYS · SNAPSHOT OF 30 OCTOBER 2026

PROTOCOL	CATEGORY	FEES 30 D	REVENUE 30 D	KEPT
Tether	Stablecoin issuer	522.4 M $	522.4 M $	100 %
Circle USDC	Stablecoin issuer	209.0 M $	209.0 M $	100 %
PumpSwap	Exchange (AMM)	105.9 M $	14.0 M $	13 %
Polymarket US	Prediction market	86.5 M $	0.0 M $	0 %

Hyperliquid Perps	Derivatives	80.9 M $	64.1 M $	79 %
Papertrade	Derivatives	67.9 M $	67.9 M $	100 %
pump.fun	Launchpad	61.9 M $	44.8 M $	72 %
Uniswap V4	Exchange (AMM)	54.9 M $	0.0 M $	0 %
Lido	Liquid staking	49.4 M $	3.0 M $	6 %
Aave V3	Lending	36.2 M $	4.6 M $	13 %
Uniswap V3	Exchange (AMM)	33.0 M $	7.9 M $	24 %
Polymarket International	Prediction market	32.4 M $	15.9 M $	49 %
Hyper Foundation HYPE Staking	Staking pool	31.9 M $	1.0 M $	3 %
Flap sh	Launchpad	29.5 M $	7.0 M $	24 %

“Fees”: what users paid. “Revenue”: the share the protocol keeps for its treasury or its token holders; the rest goes to depositors, validators or market makers.

WHERE DO THE FEES GO?

Tether: 522 M fees, 522 M revenue. The fees are the interest on the Treasury bills backing USDT.

To depositors / market makers | To the protocol and its token holders | Shared

Uniswap V4: 55 M fees, 0 revenue.

To depositors / market makers | To the protocol and its token holders | Shared

Hyperliquid (derivatives): 81 M fees, 64 M revenue.

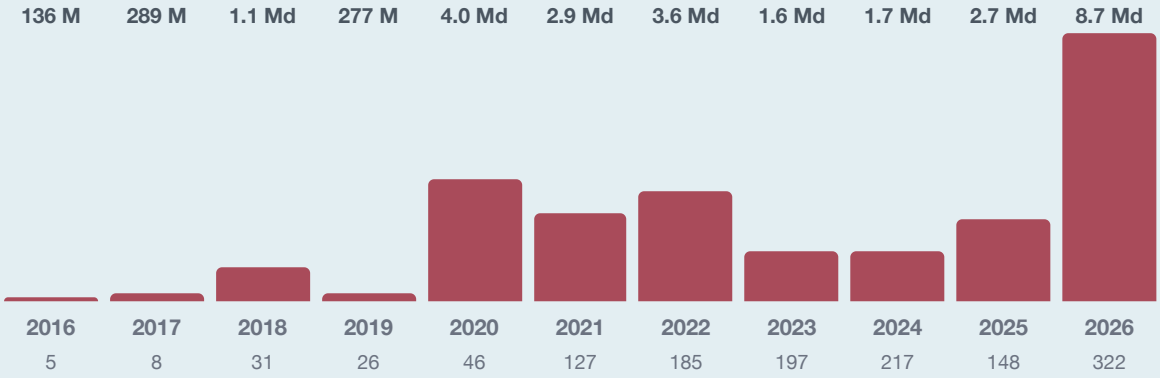
To depositors / market makers | To the protocol and its token holders | Shared

Lido: 49 M fees, 3 M revenue.

To depositors / market makers | To the protocol and its token holders | Shared

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HACKS SINCE 2016, IN MILLIONS OF DOLLARS A YEAR



TARGET	DATE	AMOUNT	HOW
Binance CEX	2026-10-29	6.36 Md $	Frontend & infrastructure · Key Leaked via Infrastructure
LuBian	2020-12-28	3.50 Md $	Key compromise · Weak Key Generation
Bybit	2025-02-21	1.40 Md $	Social engineering · Signer Phishing
Ronin Bridge	2022-03-23	624 M $	Key compromise · Validator Key Compromised
Poly Network	2021-08-10	611 M $	Access control · Improper Access Control
Binance Bridge	2022-10-06	570 M $	Bridges & cross-chain · Proof Verifier Bug
Coincheck	2018-01-26	534 M $	Key compromise · Hot Wallet Key Compromised
Mt. Gox	2014-02-07	470 M $	Key compromise · Private Key Compromised

By type, all time: Key compromise 33 % · Frontend & infrastructure 27 % · Social engineering 11 % · Access control 7 % · Bridges & cross-chain 6 % · Token & share accounting 5 % · Oracle Manipulation 3 %

1,330 incidents listed by DefiLlama, 27.6 bn $ in all. Under each bar: the number of incidents that year. On 29 October 2026 Binance lost 6.3 billion dollars in twenty

minutes: a compromised laptop, withdrawals signed with the real keys. The ledger worked exactly as designed.

What share of all money ever stolen went through **key compromise** (the “By type” line), in %?

… % Check

E · 5 MIN · FOR THE PROJECT

Three figures for your dossier

✍️ **Three figures read today on DefiLlama that you could cite in the dossier of your pension-savings app — with, for each, what it proves or what it warns of (the size of stablecoins, the share of hacks due to keys, the yield on a USDC deposit…).**

1. … (source: DefiLlama, 30 October 2026) — what it proves: …

✍️ **If your clients’ fund units were tokens: on which chain, and why that one? Three criteria drawn from what you have just seen: what it already carries, what it costs, what has happened to it.**

Chain: … Because: 1. … 2. … 3. …

DEBRIEF · 5 MIN

Three questions: what does TVL measure, and what does it not? Why do stablecoins weigh more than all of DeFi together? The largest theft in the history of crypto-assets, yesterday, hit a centralised exchange, not a contract: what does that change about the “trustless” promise?

SECTION 4.3 · BLOCK 2

🎤 TALK · 15 MIN

Lend, swap, stabilise — with no bank

The instructor's talk, or read on your own.

LENDING · AAVE-TYPE

A loan with no file: the collateral does everything

A lending protocol is a program (a *smart contract*) that holds a common pool. Nobody runs it day to day: lenders deposit tokens into it and earn interest; borrowers deposit collateral into it and borrow against that. The program knows no name, no income, no history: no score, no collection, only collateral **worth more than the loan** and a liquidation rule that applies by itself.

Below, two people who do not know each other, a lender and a borrower, use the same contract. At each step: who calls the contract, what the contract does, and the state of the pool. The figures are those of the tool in section 4.4 (LTV 80%, threshold 82.5%, bonus 5%).

▸ TWO USERS, ONE CONTRACT · STEP BY STEP

1 · Awa deposits. Awa has 10,000 USDC sitting idle. She calls *supply* on the contract: the 10,000 USDC move from her address to the pool, and the contract writes her 10,000 aUSDC in return, a receipt whose balance grows with the interest. She does not know who will borrow.

```
Pool: 10,000 USDC deposited · 0 lent · utilisation 0% · rate 0%
```

2 · Moussa deposits his collateral. Moussa has 2 ETH (3,000 USDC each) and needs 3,000 USDC for six months, without selling his ETH. He calls *supply* with 2 ETH: the contract locks them as collateral, worth 6,000 USDC according to the price oracle. He can borrow up to 80%: 4,800 USDC.

```
Moussa's collateral: 2 ETH = 6,000 USDC · maximum borrow 4,800 USDC
```

3 · Moussa borrows. He calls *borrow* for 3,000 USDC. The contract checks the rule (3,000 ≤ 4,800), records a debt of 3,000 USDC against his address and sends him 3,000 USDC: Awa's. Nobody read a file; Awa signed nothing.

```
Pool: 10,000 deposited · 3,000 lent · utilisation 30% · borrow rate 4% a year · deposit rate 1.2%
```

```
Health factor: 6,000 × 0.825 ÷ 3,000 = 1.65
```

4 · For six months. Moussa uses his USDC: he swaps them for francs, pays a supplier, or buys more ETH. Every second, the contract increases his debt at the current rate, and Awa's aUSDC balance by the share that is hers. If others borrow, utilisation rises and both rates with it.

```
After six months at 4%: debt = 3,000 × (1 + 0.04 × ½) = 3,060 USDC · Awa's aUSDC ≈ 10,060
```

5a · Repayment, if all goes well. Moussa calls *repay* with 3,060 USDC: his debt drops to zero, and *withdraw* returns his 2 ETH, which he never sold. Awa calls *withdraw*: her 10,060 aUSDC are burned and 10,060 USDC return to her address. Neither of them spoke to anything but the program.

```
Pool: 0 lent · Awa + 60 USDC · Moussa − 60 USDC, 2 ETH recovered
```

5b · The fall, otherwise. In month four, ETH drops to 1,750 USDC. Moussa's collateral is now worth only 3,500 USDC against a debt of 3,040: health factor 3,500 × 0.825 ÷ 3,040 = **0.95**. The contract does nothing by itself, it has no "sell" button: it simply makes the position liquidatable by anyone.

```
Liquidation price: 3,040 ÷ (2 × 0.825) = 1,842 USDC per ETH · ETH is below it
```

6 · A third user liquidates. A bot, anybody's, watches the health factors. It calls *liquidationCall*: it repays half of Moussa's debt, 1,520 USDC, and the contract gives it 1,520 × 1.05 = 1,596 USDC worth of ETH, i.e. 0.912 ETH, taken from the collateral. The bot earns 76 USDC; Awa had nothing to do.

```
Moussa: 1.088 ETH left · debt 1,520 USDC · factor 1.088 × 1,750 × 0.825 ÷ 1,520 = 1.03
```

7 · The outcome. Moussa keeps 1.088 ETH and still owes 1,520 USDC: the 76 USDC bonus went in a few seconds, with no notice and no call. Awa was protected without doing anything: that is what over-collateralisation and public liquidation are for, not judging Moussa. What can still go wrong: a crash faster than the bots (the collateral falls below the debt: bad debt), an oracle giving a wrong price, a bug in the contract. When many loans are liquidated at once, the bots resell the seized ETH, which pushes the price down further: the **liquidation cascade**.

✓ All 8 steps are shown

The four rules the contract applies.

- **Maximum loan-to-value (LTV).** At opening, you borrow at most 80% of the collateral's value.
- **Liquidation threshold.** If the debt exceeds 82.5% of the collateral's value, the loan becomes liquidatable.
- **Health factor.** Collateral value × threshold ÷ debt. Above 1, nothing happens; below 1, anyone can repay part of the debt in the borrower's place and take some of their collateral with a 5% **bonus**. Automatic, public, no judge.
- **The rate.** It follows the pool's **utilisation** (the share lent out): the more of the pool is lent, the higher the rate, to attract deposits and slow borrowing. The interest borrowers pay goes to the depositors, less a reserve the protocol keeps.

Health factor = collateral × liquidation threshold ÷ debt

What it is used for, in practice.

- **Liquidity without selling.** Moussa needs 3,000 USDC for six months but wants to keep his ETH: he posts it as collateral instead of selling it, and gets it back as it was. No sale, no realised gain, no file, about twenty seconds.
- **A return on idle cash.** Awa places stablecoins she is not using; the rate is public, it moves every block, and she withdraws when she wants, as long as the pool is not fully lent out.
- **Leverage.** Borrow USDC against ETH to buy more ETH, and deposit that too: the position rises faster, and gets liquidated faster.
- **What it is not.** Credit for someone who has nothing: you must already hold more than you borrow. The rate is not fixed, the contract can have a bug, and a crash faster than the liquidators leaves **bad debt** that the protocol has to absorb.

Aave's real parameters vary from one asset to another and from month to month; the ones on this page are orders of magnitude, chosen for the calculation.

THE EXCHANGE · UNISWAP-TYPE

No order book: a formula

An **automated market maker** (AMM) is a pot holding two tokens, for example 100 ETH and 300,000 USDC. The program accepts any swap that leaves the product of the two quantities unchanged: **x × y = k**. The price is the ratio of the two reserves (300,000 ÷ 100 = 3,000 USDC per ETH). The more you buy at once, the more you empty the pot on one side and the higher the average price: that is **price impact**. Each swap pays a fee (often 0.3%) to the pot's depositors. If the pot's price drifts from the market's, arbitrageurs bring it back by swapping — and earn the gap.

Received = y × Δx × (1 − fee) ÷ (x + Δx × (1 − fee))

STABLECOINS

A token worth one dollar — provided that…

- **Backed by reserves.** A company issues one token for each dollar received and places the dollars in cash and Treasury bills (Circle's USDC, Tether's USDT). The value holds as long as the reserves exist and can be reached. In March 2023, 3.3 billion dollars of USDC's reserves were stuck at Silicon Valley Bank: USDC fell below 0.90 dollar for a weekend.
- **Backed by crypto-assets.** DAI is created by borrowers who deposit more collateral than they issue, with automatic liquidation — Aave's mechanism, applied to a currency.
- **Algorithmic.** No reserves: the peg holds through a swap mechanism with a second token. TerraUSD (UST) worked this way; in May 2022 a wave of selling sent both tokens to almost zero, and tens of billions of dollars vanished within days.

▸ WHAT BACKS THE VALUE OF THIS TOKEN?

USDC (Circle)

Currency reserves **Crypto-asset collateral** **Algorithm**

USDT (Tether)

Currency reserves **Crypto-asset collateral** **Algorithm**

DAI

Currency reserves **Crypto-asset collateral** **Algorithm**

TerraUSD (UST), until May 2022

Currency reserves | Crypto-asset collateral | Algorithm

EURC (Circle), a euro token

Currency reserves | Crypto-asset collateral | Algorithm

0 of 5 sorted

SECTION 4.4 · BLOCK 2

WORKSHOP · 50 MIN

DeFi labs · borrow, swap

A · 15 MIN · THE LOAN

2 ETH as collateral, USDC borrowed, and the ETH price falling

First write your prediction: at what ETH price will this loan be liquidated? Then move the price. Maximum LTV 80%, liquidation threshold 82.5%, liquidation bonus 5%.

ETH deposited as collateral

2

USDC borrowed

3000

ETH price, in USDC

6,000	4,800	1.65	1,818.18
COLLATERAL VALUE, USDC	MAXIMUM LOAN AT 80%	HEALTH FACTOR	ETH LIQUIDATION PRICE

Healthy. The ETH price can still fall by 39.4% before liquidation.

2 ETH at 3,000 USDC as collateral, 3,000 USDC borrowed, liquidation threshold 82.5%. The **health factor**, to two decimals?

… Check

At what **ETH price** does this loan become liquidatable? Rounded to the nearest unit.

… USDC Check

At 1,700, a bot repays 1,500 USDC of the debt, with a 5% bonus. How much **bonus does the borrower lose**, in USDC?

… USDC Check

B · 15 MIN · THE SWAP

A pot of 100 ETH and 300,000 USDC, and your order

Choose the direction and the amount. Compare the average price obtained with 3,000: the gap is the price impact, plus the fee.

Direction

buy ETH with USDC

Amount paid in (USDC or ETH)

30000

9.0661	3,309.03	10.30 %	3,629.01
ETH RECEIVED	AVERAGE PRICE OBTAINED, USDC PER ETH	GAP WITH 3,000 (IMPACT + FEE)	POT PRICE AFTER THE SWAP

Sell **10 ETH** into the pot (100 ETH, 300,000 USDC, 0.3% fee). How many USDC do you receive, to the nearest unit?

… USDC Check

Buy ETH with **30,000 USDC**. How many ETH do you receive, to two decimals?

… ETH Check

C · 15 MIN · THE DEFI CAPSTONE

A complete case, and the module certificate

The DeFi capstone ↗ chains deposit, borrowing, swap and liquidation on a single wallet, and issues a certificate at the end. The course "How DeFi works" ↗ goes over the theory if needed.

The DeFi capstone, through to the certificate.

Finished, certificate obtained | Started | Not yet

1 of 1

✍ **Would you place your customers' pension savings in a DeFi lending pot paying 4%? Three reasons, each taken from a mechanism seen this morning.**

e.g. No: 1) the yield comes from borrowers who can be liquidated in a cascade… 2) … 3) …

SECTION 4.5 · BLOCK 2

✅ QUIZ · 10 MIN

End-of-morning quiz

INTERACTIVE QUIZ · COUNTS TOWARDS PARTICIPATION

Eight questions, one attempt each, all explained

Launched in class; this page unlocks by itself. On your own, without an AI assistant. A missed quiz scores zero, unless arranged in advance.

🔒 **The quiz starts together.** It is launched in class; this page unlocks by itself within seconds, without reloading.

0 / 8 answered

1. **A transaction is signed with Awa's private key. This proves…**

- that Awa is its author, whatever happens
- that the holder of the private key authorised it, and that it has not been altered
- that the transaction is finalised
- that the recipient is identified

2. **Why can a transaction in an old block not be altered quietly?**

- Because the blocks are encrypted
- Because FINMA keeps a copy
- Because each block contains the previous one's fingerprint: everything after it would change, and the network would reject it
- Because transactions are anonymous

3. **A simple transfer uses 21,000 gas, at a price of 2 gwei. The fee is…**

- 0.000042 ETH
- 0.042 ETH
- 42 ETH
- 21,000 ETH

4. A DeFi loan has a health factor of 1.65. If the collateral's price falls by 40%, the factor becomes…

- 1.25: healthy
- 0.99: liquidatable
- 1.65: it does not depend on the price
- 0.40

5. On an x × y = k AMM, buying a lot of ETH at once…

- costs the quoted price, plus 0.3%
- costs a higher average price, because each ETH removed makes the next one dearer
- lowers the ETH price in the pot
- is refused beyond 10% of the pot

6. In March 2023, USDC fell below 0.90 dollar for a weekend. Why?

- A bug in its smart contract
- Part of its reserves was stuck at the failed Silicon Valley Bank
- Its algorithm gave way, like TerraUSD's
- The United States had banned it

7. According to Wüst and Gervais's tree, if an always-available trusted third party can keep the ledger…

- a public blockchain is needed
- a private blockchain is needed
- a blockchain is not needed: the third party does it better, faster and cheaper
- a stablecoin is needed

8. In a "sandwich" attack, the bot makes money because…

- it knows your private key
- your order is visible in the public queue before it is executed, and it places a buy in front and a sell behind
- it hacks the AMM's contract
- it is always the validator of the next block

SECTION 4.6 · BLOCK 3

🎤 TALK · 20 MIN

Tokenisation and central bank digital currencies

The instructor's talk, or read on your own.

TOKENISING

A token is a ledger entry that represents a right

Tokenising a building, a bond or a fund unit means recording the ownership right in a shared ledger rather than in a custodian's. What you gain: transfers settled in minutes instead of two days, smaller units, rules (who may buy, when the coupon is paid) written into the token. What you do not gain: **the blockchain does not know that the building exists**. The link between the token and the asset remains a contract, a company, a custodian and an applicable law — exactly as for a share.

SWITZERLAND

A law, an exchange, a wholesale currency

- **The DLT Act** (the federal act adapting federal law to developments in distributed ledger technology, in force in 2021) creates **ledger-based securities** (art. 973d CO): a security that exists only in an electronic register, transferable on that register with the same legal certainty as a paper security. It also creates a licence for a **DLT trading facility** and protects clients' crypto-assets if the custodian goes bankrupt.
- **SDX** (SIX Digital Exchange), the SIX group's digital exchange, authorised by FINMA in 2021, issues and settles digital bonds.
- **Helvetia**: the SNB makes a **wholesale central bank digital currency** available to banks on SDX's platform, to settle tokenised bonds in central bank money. In pilot operation since December 2023, extended by the SNB until at least June 2028. A digital currency **for the public** (retail) is not planned by the SNB.

WAEMU

Studies, and no decision

- **A BCEAO digital currency?** On 22 July 2026, presenting its annual report, the BCEAO said that a study phase is under way and that **no deployment decision has been taken**. The “eCFA” launched in Senegal in 2016 was not central bank money: it was electronic money issued by a commercial bank, on a start-up’s technology.
- **Crypto-assets.** The BCEAO points out that they are not money and are not regulated in WAMU; a regulatory framework is being prepared. Neighbouring Nigeria launched a retail digital currency, the eNaira, in 2021: its adoption has remained very low.

CENTRAL BANK DIGITAL CURRENCY: RETAIL, WHOLESALE, OR NOT AT ALL?

Helvetia: digital francs from the SNB to settle tokenised bonds on SDX

Retail CBDC | Wholesale CBDC | Not a CBDC

The Central Bank of Nigeria’s eNaira (2021)

Retail CBDC | Wholesale CBDC | Not a CBDC

The “eCFA” launched in Senegal in 2016

Retail CBDC | Wholesale CBDC | Not a CBDC

USDC

Retail CBDC | Wholesale CBDC | Not a CBDC

The digital euro being prepared at the ECB

Retail CBDC | Wholesale CBDC | Not a CBDC

A TWINT payment

Retail CBDC | Wholesale CBDC | Not a CBDC

0 of 6 sorted

SECTION 4.7 · BLOCK 3

🛠 WORKSHOP · 55 MIN

Split a building · do you need a blockchain?

A · 20 MIN · THE FRACTIONALISATION SIMULATOR

A building in Lausanne, in 1,000-franc tokens

A company owns the building; its equity is split into tokens. Change the value, the debt, the rent: what one token is worth, what it yields — and what it loses when the building loses 10%.

Building value, CHF

12000000

Mortgage, CHF

4000000

Number of tokens

8000

Rent net of costs, CHF per year

360000

Mortgage rate, %

2

1,000 CHF	35.00 CHF	850 CHF
VALUE OF ONE TOKEN	PAID OUT PER TOKEN PER YEAR (3.50%)	ONE TOKEN IF THE BUILDING LOSES 10% (-15.0%)

Building 12 million, mortgage 4 million at 2%, net rent CHF 360,000, 8,000 tokens. How much does one token **pay out per year**, in CHF?

… CHF Check

The building loses 10%. What is **one token** worth, in CHF?

… CHF Check

✍ **Could your customers' fund units be tokens (ledger-based securities)? What the customer would gain, and what would not change.**

e.g. Gained: the customer sees their unit in a public register… Unchanged: a Swiss custodian, the fund and the exchange rate are still needed…

B · 20 MIN · BY TEAM · “DO YOU NEED A BLOCKCHAIN?”

Six questions, applied to your app

The method is published: Wüst and Gervais, *Do you need a Blockchain?* ↗ (2018). Take one precise function of your app — the register of customers’ units, the monthly statement, the transfer of payments from Africa — and answer for it. Start again for a second function.

The function of your app you are testing

e.g. the register of custo

Do you need to store data (a state: balances, a register)?

Yes | No

✍ **For each function tested: the tree’s result, and your decision argued in two sentences. This is the dossier’s “argued technology choice”.**

The tree inserts its result; add your decision.

DEBRIEF · 15 MIN

Each team reads its two results. Where did the tree say “no blockchain”? What tipped the answer — an available trusted third party, known actors? And if the answer is “no”, what from today is still useful for your app? Section 4.9 gives part of the answer.

SECTION 4.8 · BLOCK 4

🎤 TALK · 15 MIN

MEV, three hacks, MiCA

The instructor's talk, or read on your own.

THE "SANDWICH" ATTACK

Your order is public before it is executed

A transaction waits in the **public queue** (the *mempool*) before it enters a block, and whoever builds the block chooses the order. A bot sees your purchase of 30,000 USDC of ETH on the pot from section 4.4. It **buys just before you** (the price rises), your purchase goes through at the higher price, and it **sells just after**. It gains exactly what you lose, within your **slippage tolerance**: with 1%, you agree to receive 1% less than expected, and that is what the bot takes. This gain that the order of transactions makes extractable is called **MEV** (*maximal extractable value*).

In this example the bot puts 1,586 USDC in front of you, you receive 8.9754 ETH instead of 9.0661, and it sells back with **a 318 USDC gain**, pot fees included. The simulator "The dark forest" ↗ shows it block by block.

With no attack, 30,000 USDC buy 9.0661 ETH. With a 1% slippage tolerance, what is the **minimum ETH** you agree to receive, to two decimals?

… ETH Check

THREE HACKS

The code, the keys, the screen

- **The DAO, June 2016 — the code.** An investment fund run by a smart contract. A flaw allowed a withdrawal to be requested again before the contract had updated the balance (*re-entrancy*): about 3.6 million ETH diverted. Ethereum rewrote history through a split (*hard fork*); those who refused it carried on under the name Ethereum Classic. Lesson: "code is law" applies to bugs too.
- **Ronin, March 2022 — the keys.** The bridge of the game Axie Infinity required 5 signatures out of 9 validators. The attackers obtained 5 keys and withdrew about 625

million dollars; the FBI attributed the attack to North Korea. Lesson: a secure chain does not protect badly kept keys.

- **Bybit, February 2025 — the screen.** The platform signed its transfers from a multi-signature “cold” wallet. The signing interface had been hacked: the signers approved what they saw on screen, not what they were signing. About 1.5 billion dollars of ETH, the largest theft in crypto-asset history, also attributed to North Korea. Lesson: check what you sign, on a second device.

▸ THE CODE · THE KEYS · THE SCREEN: WHERE DOES EACH RISK COME FROM?

The DAO (2016): a withdrawal requested again before the balance was updated

The code | The keys | The screen, the person

Ronin (2022): 5 validator keys out of 9 in the attackers’ hands

The code | The keys | The screen, the person

Bybit (2025): signers approve what the interface shows them, not what they sign

The code | The keys | The screen, the person

A student gives their secret phrase to a fake “MetaMask support”

The code | The keys | The screen, the person

A platform keeps all its customers’ keys on one server connected to the Internet

The code | The keys | The screen, the person

0 of 5 sorted

THE FRAMEWORK

MiCA in the EU, FINMA in Switzerland, the BCEAO

- **European Union · MiCA** (Regulation 2023/1114). Since 30 June 2024 for stablecoins, since 30 December 2024 for the rest: an issuer of a stablecoin backed by a single

currency (an *e-money token*) must be a credit or e-money institution, with segregated reserves and a right to redemption at par; a provider (platform, custodian, adviser) needs an authorisation, valid across the Union. The national transitional periods ended on 1 July 2026 at the latest.

- **Switzerland.** No MiCA: FINMA applies existing law according to the token's function (payment, investment, utility), the DLT Act, and the AMLA to every provider that transfers crypto-assets on behalf of others.
- **WAEMU.** No framework specific to crypto-assets so far (see section 4.6).

SECTION 4.9 · BLOCK 4

PROJECT AND GAME · 75 MIN

Prove the money is there · DeFi Millionaire

A · 15 MIN · THE PROOF OF INCLUSION

Each customer checks that their units are counted, without seeing anyone else's

After FTX's bankruptcy (November 2022), platforms began publishing a "proof of reserves". Its "liabilities" half is a **Merkle tree**: the fingerprint of each account (identifier, balance, and a secret *salt* that prevents guessing the others' balances) is combined two by two up to a single fingerprint, the **root**, published on the same day for everyone. Each customer receives their path — three fingerprints here — and recomputes the root themselves. If their balance had been left out or reduced, the root would not match. Here are eight customers of the app, in fund units.

› THE MERKLE TREE OF THE EIGHT CUSTOMERS · SHA-256 COMPUTED IN YOUR BROWSER

You are customer: c03

Root published by the company: 3ab7213e6ca36165…

Check my inclusion | What if the company had reduced my balance?

The root proves that your balance is *counted*. It does not prove that the units exist: that is the other half — an attestation from the custodian and the auditor that the assets held cover the sum of the balances. A blockchain can serve to **publish the root** with a timestamp and unalterably; it replaces neither the custodian nor the auditor.

B · 30 MIN · BY TEAM · THE PROJECT

Prove that your customers' money is there

1. **What your customer can check themselves, without taking your word for it: what, by what means (custodian's statement, published root, token register…), how often — and who attests the other half (the assets).**

e.g. Every month the Merkle root of the balances is published… ; the Swiss custodian attests… ; the auditor…

2. The lesson of the three hacks for your app: who can move customers' money, with how many signatures, on which devices — and what happens if an employee or a phone is compromised.

e.g. No employee can move the units: only the custodian, on instruction… ; two signatures out of three, checked on a second device…

C · 25 MIN · DEFI MILLIONAIRE, BY TEAMS

The class game DeFi Millionaire ↗ is played on the room's screen, both teams at once. The questions go back over the day.

DeFi Millionaire 24:56 Pause ↺

BEFORE DAY 5 · SATURDAY 7 NOVEMBER

- Day 5 covers artificial intelligence, data, regtech and compliance: a fraud-detection notebook (Colab), a KYC/AML file, the "Regulator versus fintech" game.
- For the project: note which regulator, in your customer's country and in Switzerland, would have to authorise your app — Day 5 will have you write the regulatory dossier.